Combined Mean:

Let $\overline{x}_1$ and $\overline{x}_2$ are the arithmetic mean of two groups (having the same unit of measurement of a variable), based on n1 and n2 observations respectively. Then the combined mean can be calculated using

$$\text{Combined Mean} = \bar{x}_{12} = \frac{n_1\overline{x}_1 + n_2\overline{x}_2}{n_1 + n_2}$$

Remark : The above result can be extended to any number of groups.

Example 5.6

A class consists of 4 boys and 3 girls. The average marks obtained by the boys and girls are 20 and 30 respectively. Find the class average.

Solution:

$n_1 = 4,\ \overline{x}_1 = 20,\ n_2 = 3,\ \overline{x}_2 = 30$

$$\text{Combined Mean} = \bar{x}_{12} = \frac{n_1\overline{x}_1 + n_2\overline{x}_2}{n_1 + n_2}$$

$$= \left[\frac{4 \times 20 + 3 \times 30}{4 + 3}\right]$$

$$= \left[\frac{80 + 90}{7} = \frac{170}{7}\right] = 24.3$$

5.3.2 Geometric Mean(GM)

(a) G.M. For Ungrouped data

The Geometric Mean (G.M.) of a set of n observations is the nth root of their product. If $x_1, x_2, \ldots, x_n$ are n observations then

$$\text{G. M.} = \sqrt[n]{x_1, x_2 \ldots x_n} = (x_1 . x_2 \ldots x_n)^{\frac{1}{n}}$$

Taking the nth root of a number is difficult. Thus, the computation is done as under

$$\log \text{G.M.} = \log (x_1 . x_2 \ldots x_n)$$

$$= (\log x_1 + \log x_2 + \ldots + \log x_n)$$

$$= \frac{\sum_{i=l}^{n} \log x_i}{n}$$

$$\text{G.M.} = \text{Antilog} \frac{\sum_{i=l}^{n} \log x_i}{n}$$

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